

# Bitwise differential trail through **TWINE** found by SAT

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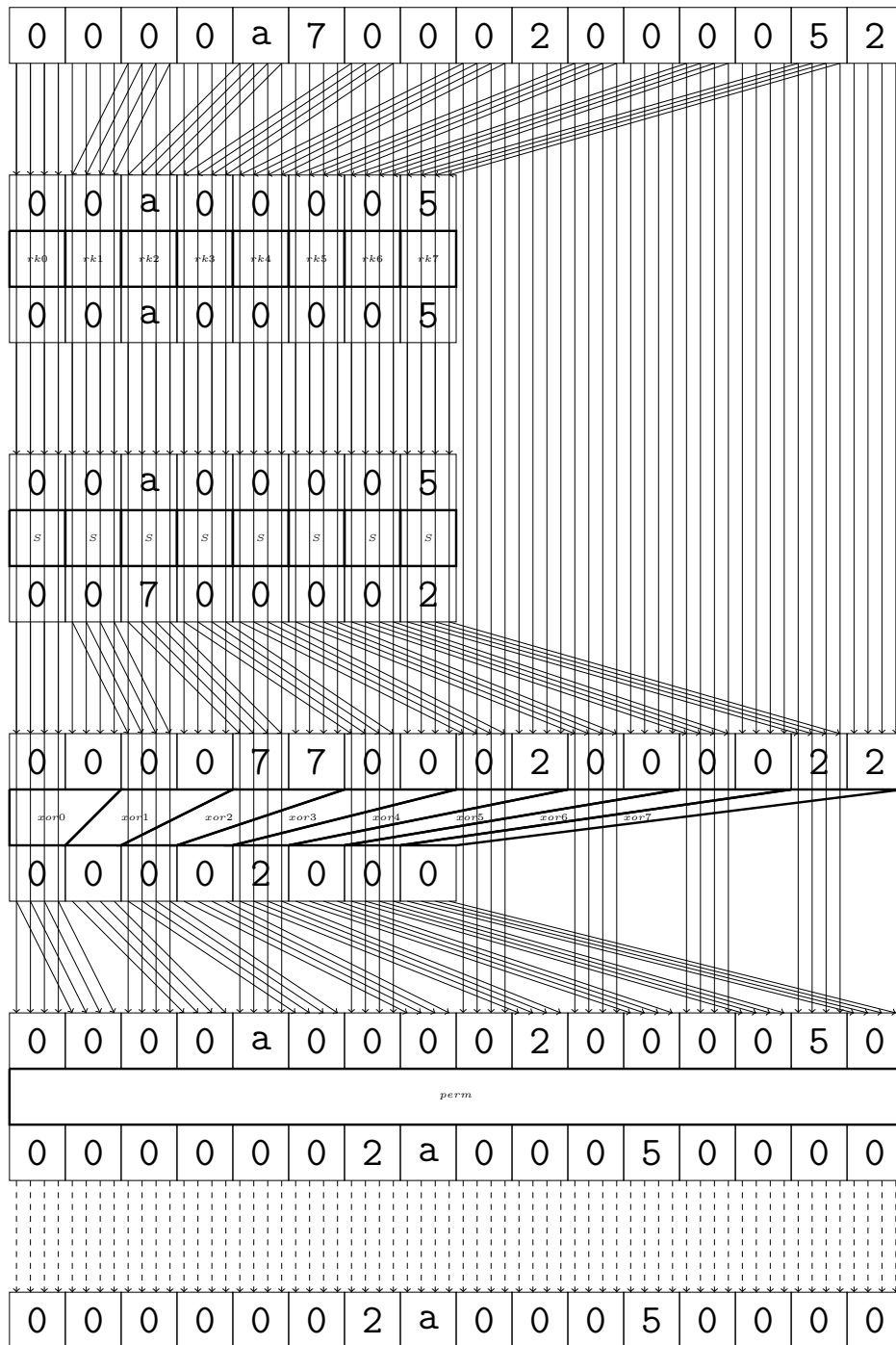
# 1 TWINE

Maximal differential probability:  $2^{-16}$

|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0     | 0 | 0 | 0 | a | 7 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 5 | 2 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | a | 7 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 5 | 2 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 2 | a | 0 | 0 | 0 | 5 | 0 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 2 | a | 0 | 0 | 0 | 5 | 0 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 5 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 5 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 5     | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 5     | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2     | 0 | 0 | 0 | 0 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2     | 0 | 0 | 0 | 0 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| a     | 0 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 5 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| a     | 0 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 5 | 0 | 0 | 0 |
| final |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| a     | 7 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 5 | 2 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| a     | 7 | 0 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 5 | 2 | 0 | 0 |

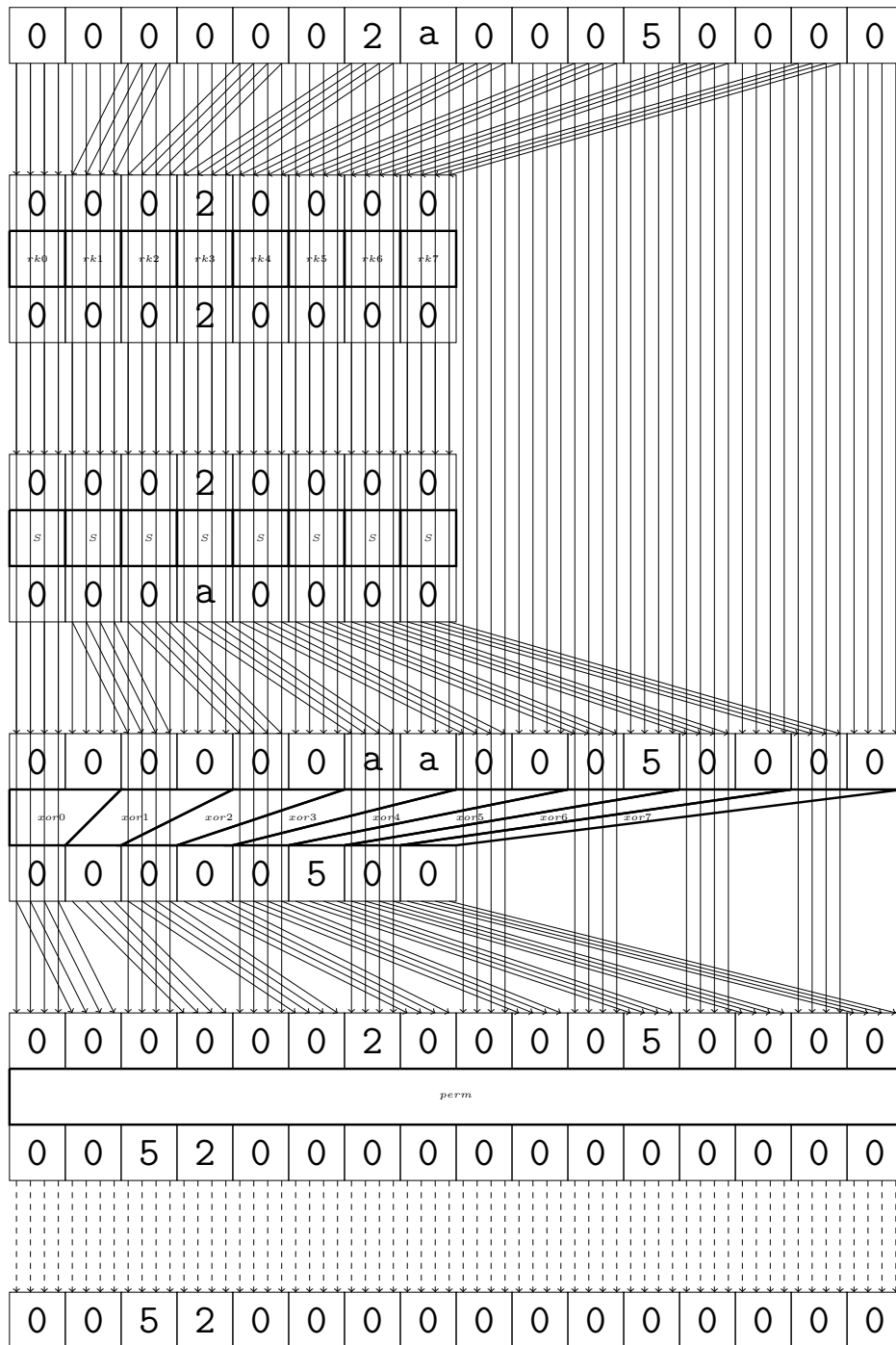
## 2 round

Maximal differential probability:  $2^{-4.0}$



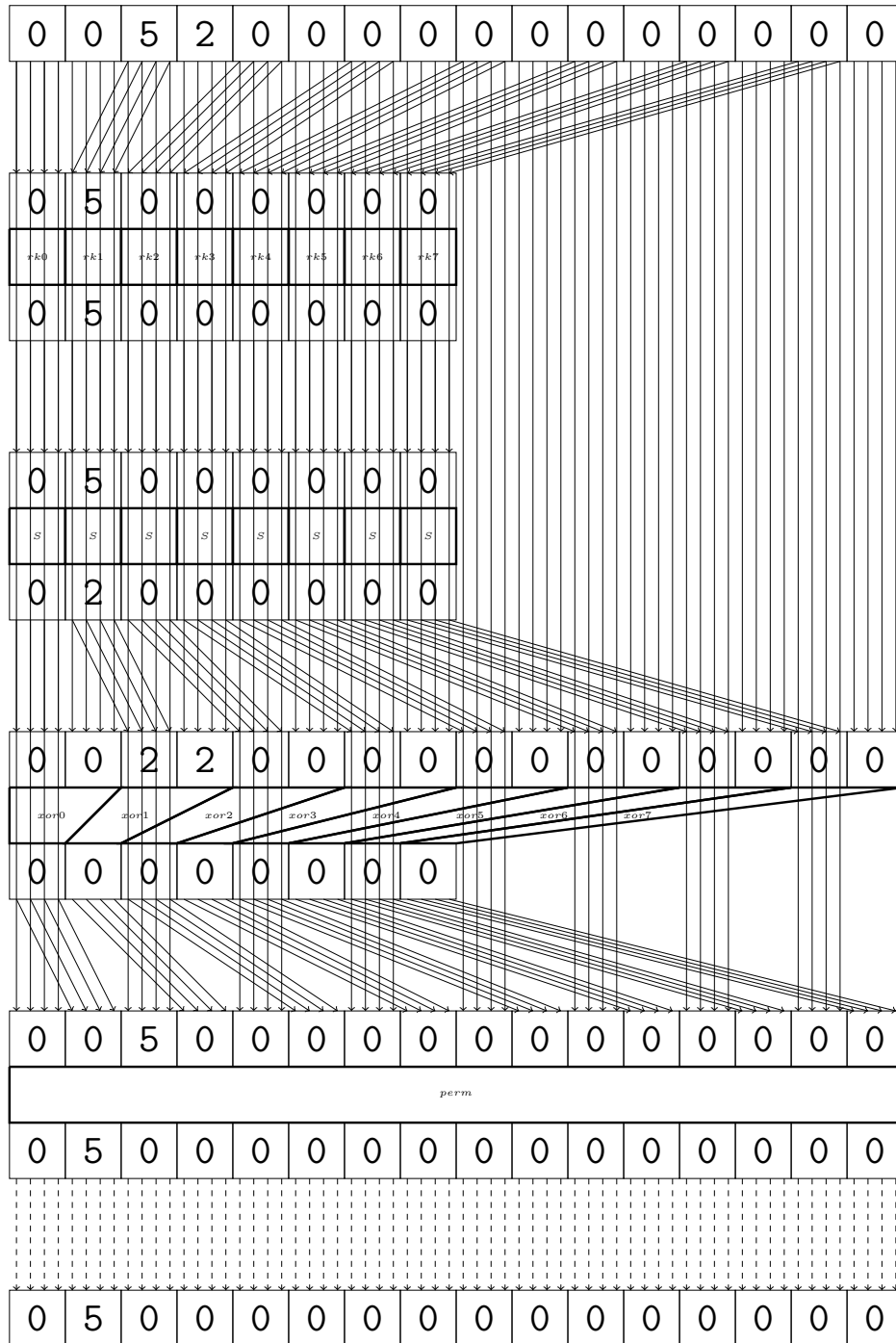
### 3 round

Maximal differential probability:  $2^{-2.0}$

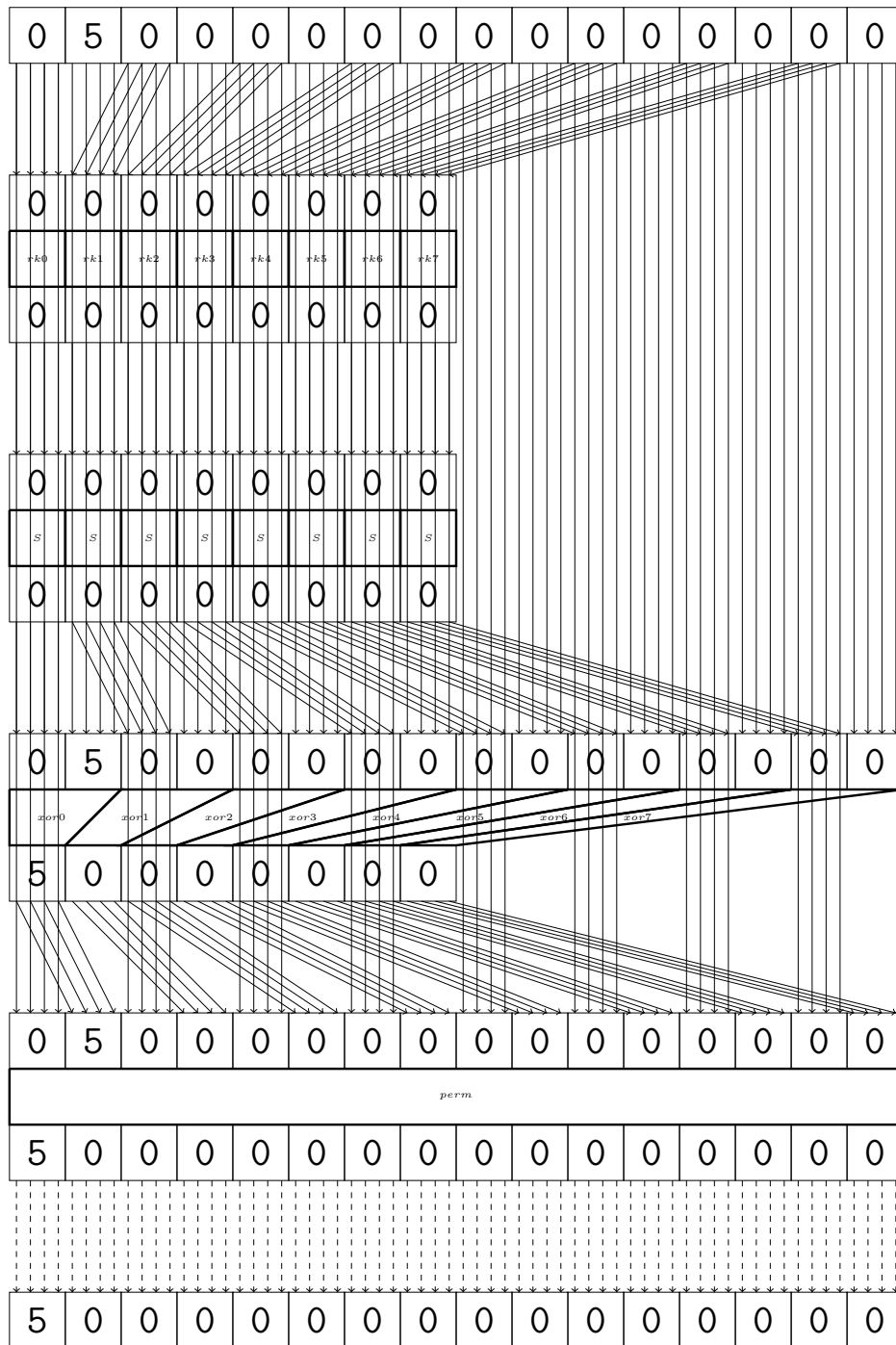


## 4 round

Maximal differential probability:  $2^{-2.0}$

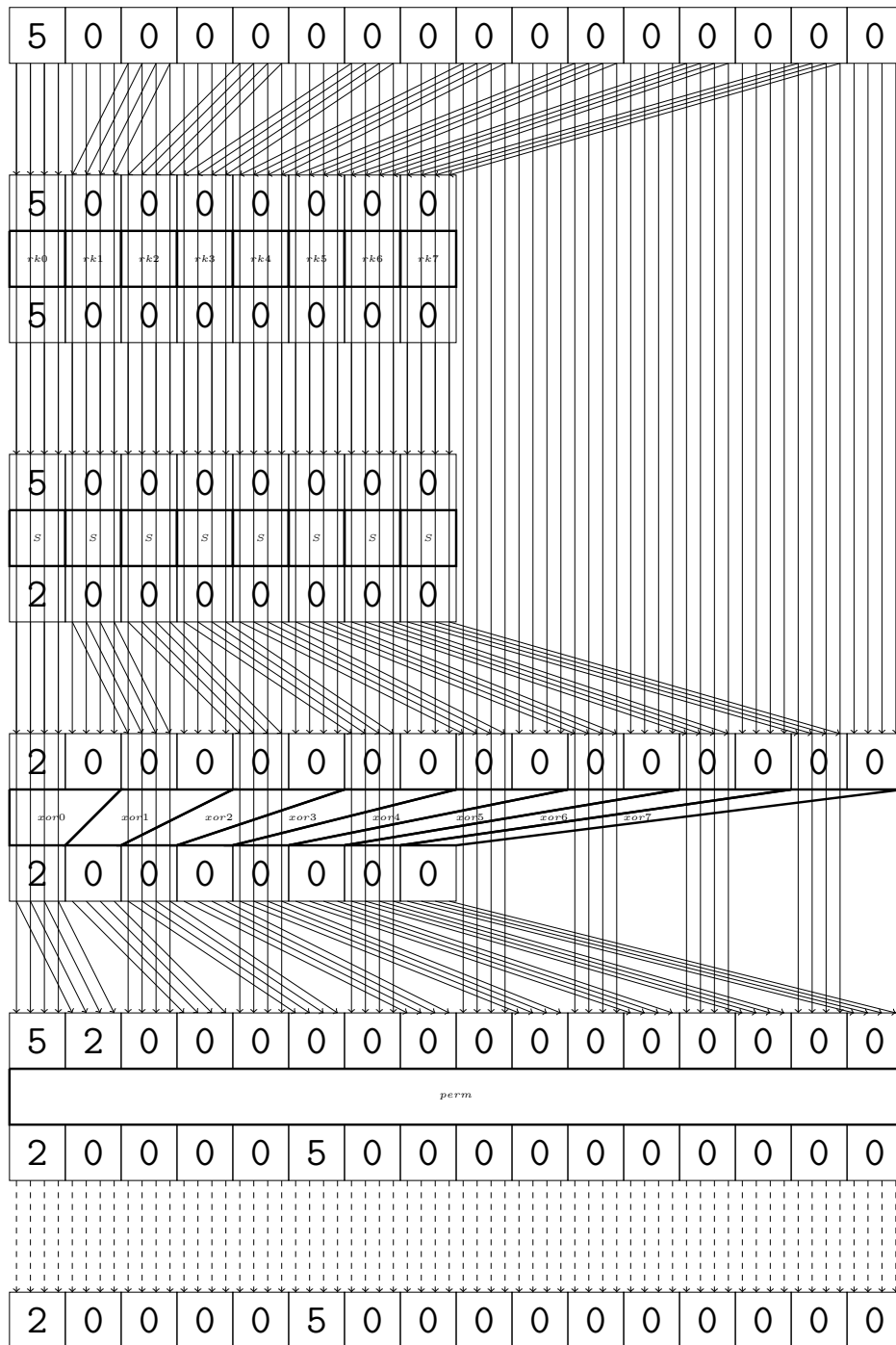


## 5 round

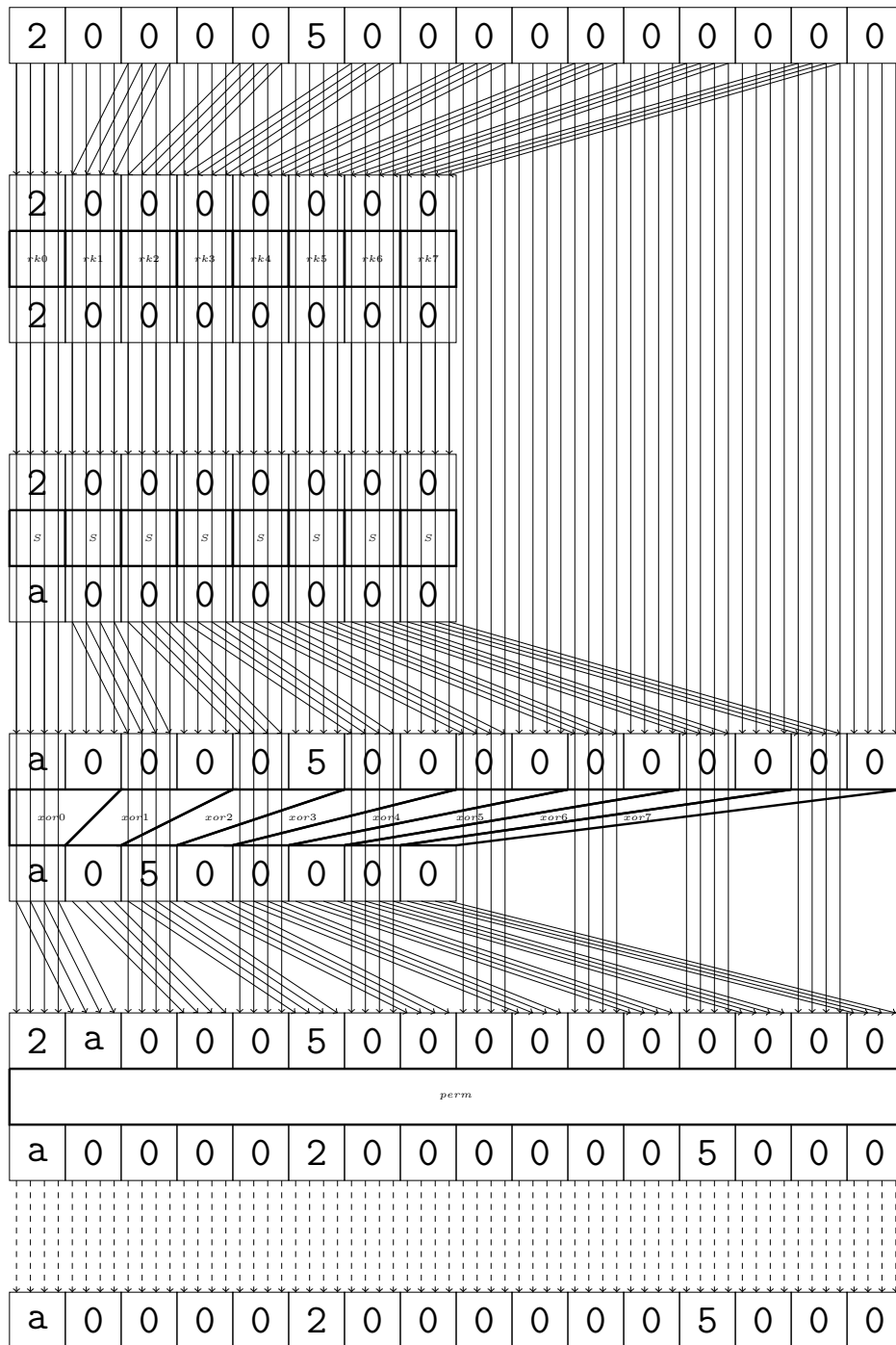
Maximal differential probability:  $2^{-0.0}$ 

## 6 round

Maximal differential probability:  $2^{-2.0}$



## 7 round

Maximal differential probability:  $2^{-2.0}$ 



## 8 final

Maximal differential probability:  $2^{-4.0}$

